



FACULTY OF INFORMATION TECHNOLOGY

ICS 1204: INTEGRAL CALCULUS

DATE: March 18, 2016

TIME: 2 Hours

INSTRUCTIONS

Answer question ONE and any other two

Question One (30 marks)

a) Evaluate the following definite integrals

i. $\int_2^3 x^2 \sqrt{2x^3 - 4} dx$ (5 marks)

ii. $\int_1^2 x^3 \ln x dx$ (5 marks)

iii. $\int_0^{\pi} \sin^2 \theta d\theta$ (5 marks)

b) Find the area enclosed by the curves $y = x^2$ and $y = 2 - x^2$ (5 marks)

c) Find the trapezoidal and Simpson's approximation $\int_0^{10} v(t) dt$ where v is the given tabulated function.

t	0	1	2	3	4	5	6	7	8	9	10
$v(t)$	12	14	17	21	22	21	15	11	11	14	17

d) Find the volume of revolution generated by revolving about the x axis the arc $y = e^x$ from $x = 0$ to $x = 1$. (6 marks)
(4 marks)

Question two (20 marks)

- a) i. Obtain the reduction formulae for $I_n = \int \cos^n x \, dx$ (6 marks)
- ii. Hence evaluate $\int_0^{\pi} \cos^3 x \, dx$ (4 marks)
- b) Calculate the trapezoidal and Simpson's approximation to $\int_0^2 \sqrt{1+x^3} \, dx$ with $n = 5$. (10marks)

Question three (20 marks)

- a) Solve the following differential equations $\frac{dy}{dx} = 2x(1+y)$. Hence find a particular solution if $x = 0$ when $y = 2$ (10 marks)
- b) A radioactive substance decays at a rate which is proportional to the amount of the substance present. Given that the half life is 1500 years and there is originally 400kg of the substance, find how much will remain after 100 years. (10 marks)

Question Four (20 marks)

- a) Resolve $\frac{3x-5}{(x+3)(x-2)}$ into partial fractions. Hence evaluate $\int_3^4 \frac{3x-5}{(x+3)(x-2)} \, dx$ (8 marks)
- c) Find the volume of the solid generated by revolving the region bounded by $y = 4 - x^2$, $x = 0$, $y = 0$ about the x axis. (6 marks)
- d) Evaluate the following definite integral $\int_0^{\pi/2} x \cos x \, dx$ (6 marks)

Question Five (20 marks)

- a) Find the Maclaurine series for the function $f(x) = e^x \cos x$ (7 marks)
- b) Expand $f(x) = \ln x$ as a series of ascending powers of $(x-a)$ up to the term containing $(x-a)^3$ and use it to evaluate $\ln 1.01$ (7 marks)
- c) Given that $y = x^{3/2}$. Find the length of the arc of the given curve between $x = 0$ to $x = 1$. (6 marks)

